Executable Verification and Analytic Repair in LLM-Guided Quantum Circuit Search: What a Controlled Benchmark Does, and Does Not, Show

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Abstract

We ask what an executable verifier and an analytic repair step actually change when a large language model (LLM) turns a *symmetry hypothesis* into an *executable* quantum predictor, under matched evaluation budgets. The mathematical tools we use (group twirls, channel and predictor equivariance residuals, an orthogonal-projection risk identity) are established; the object of study is behavioural and measurable: the gap between *proposing* a transformation, *implementing* it, and *performing*. Every claim below is traceable to a frozen revised protocol, a corrected analysis, or a bibliographic record; the change history is in Appendix D rather than in the narrative. Four things survive measurement. (i) A self-audit of our own prior LLM-evolutionary study: its selection score used test data, one architecture-wide equivariance claim is false (measured channel defect 0.0597), and significance was computed at proposal rather than run level. (ii) A causal repair experiment on a ring benchmark whose replication units, controls and floors are stated explicitly: at η=0 the target is direction-free, so the previous 24-row record contains 8 units, not 24 (recomputed: interval $[-0.00217, -0.00002]$, sign-flip p=0.125); the redesigned experiment with 24 genuine η=0 units finds that the repaired arm beats the original (-0.00134, Holm p=0.043) but *not* the gate-count-matched duplicate control (-0.00060, CI $[-0.00135, +0.00019]$), so the symmetry-specific benefit is not established, while the η=1 cost of enforced invariance is large, robust and budget-increasing ($+0.03605$, CI $[+0.02998, +0.04210]$ under the primary estimand at 200 steps, rising to $+0.057$ at 1800). (iii) In the frozen 120-run search study, the endpoint that had been omitted — inference-time Reynolds averaging of the deployed predictor — is the one effect that is consistent and large (-0.0486, CI $[-0.0708, -0.0284]$, 7/12 runs eligible on `graphs_n6`), whereas verifier *feedback* changes development-selected risk by an undetected amount; the same averaging applied to non-LLM arms also helps, so the gain is not attributable to the LLM (we do *not* establish that the two are equal). On the held-out endpoint that had been missing, the same repair step gives -0.02846 (CI $[-0.04704, -0.01190]$, 7/12 runs) on `graphs_n6`; on fresh `ring_n6` inputs the deployed gate fires on 0/12 runs, so deployed and candidate coincide there. (iv) Verification as implemented does not identify the group: 10 of 299 graph proposals carried the task's declared group, and the other 289 are *valid* symmetry hypotheses of proper subgroups — none violated label preservation, and none could have, because every permutation of six vertices preserves that target's labels, so the gate cannot fail there by construction. On the ring, where the scored target admits only 12 of 720 permutations, the same gate rejected 95 of 102 hypotheses that *were* symmetries of the scored target, with 0 false accepts: it is an under-sensitive soundness filter, not an identification test. (v) Treated as a *decision rule* rather than a tool, the gate is a recall bottleneck and not a safety device: averaging over the supplied group never harmed any of the 120 frozen candidates, the implemented gate forfeits 100% of

the available benefit on the ring (realised 0.00000 against −0.00345), a verifier-free defect gate recovers 79% of it (−0.00274) at precision 45/45 with no candidate harmed, and a mismatched group costs +0.00399 — the one regime in which a gate earns its keep. A hand-built equivariant architecture beats every search arm on the ring (+0.0060 to +0.0097 worse for the arms), and a one-threshold classical rule beats every arm on *development validation* on graphs (0.351 vs 0.538–0.566) but not on matched held-out rows (0.578 vs 0.501–0.545) — a claim of ours that the matched-row comparison corrected. All endpoints are labelled development, sealed or fresh; we do not claim held-out generalisation from development risk, we do not claim quantum advantage, and we record two provenance defects in our own earlier run records rather than quietly fixing them.

1 Introduction

Symmetry is a first-class inductive bias in quantum machine learning (QML): a circuit whose blocks commute with a representation of the data's symmetry group, combined with an equivariant encoding, an invariant initial state and a compatible readout, yields invariant predictions. A tempting inference is that an LLM asked to design such circuits will both *state* the right symmetry and *build* it, and that giving the model an executable check will close the gap.

This paper treats that inference as measurable, and separates three levels that casual "symmetry-aware LLM" claims conflate: (i) **proposing** a transformation (a stated group hypothesis); (ii) **implementing** it (the circuit actually commutes with the representation, at the parameters that were trained); and (iii) **performing** (the trained predictor is useful on data that no part of the pipeline adapted to). Our thesis is that the interesting, testable phenomenon is the *gap* between these levels, and that measuring it requires an endpoint for each level — not one endpoint read as three.

How this version differs. The manuscript was reviewed and found to contain identifiable validity problems: duplicated experimental units in the headline repair result, a search endpoint that did not measure the intervention it credited, generalisation language attached to development data, an unjustified "differs only in symmetry" claim, an "if and only if" claim, a mis-stated risk floor, uncalibrated p-values, a prose/table contradiction, and two materially wrong references. All were checked against the code and the primary records; all were real. Section 6 states the revised protocol frozen before any new result existed, and Appendix C records the dated chain, the model identity, the compute budget, and two provenance defects we found in our own earlier run records.

Contributions.

1. **A self-audit** (§4) of the prior in-house study this project continues [9]: test data inside the selection score, a refuted architecture-wide equivariance claim, and proposal-level significance.
2. **A causal repair experiment with explicit units, controls and floors** (§5, §7.1): $y_\eta = h + \eta b$; an operator-matched asymmetric control in addition to a gate-count-matched one; a second asymmetric control; optimiser-budget sensitivity; and a function-space evaluation on orbit-completed data that separates the population floor, its Monte-Carlo estimate, and the exact finite-orbit floor.
3. **A reanalysis of the frozen 120-run search study that restores the missing endpoint** (§6, §7.2): the paired effect of inference-time averaging on the frozen candidate, the deployed-pipeline

Table 1: Contribution-to-evidence table. "Evidence" names the executed artefact; "Does not show" is the boundary we refuse to cross.

Claim	Evidence	Does not show
Prior selection leaked test data	`research/audit.py` executes the evaluator's own AST; changing every test-derived field leaves the v2 score unchanged	that the historical task had no signal; it had a density shortcut
Enforcing equivariance changes risk causally	WP-REPAIR-2, 8 directions × 24 optimiser seeds, gate-count- *and* operator-matched controls (Table 4): robust cost at $\eta>0$, *no* established symmetry-specific benefit at $\eta=0$	an "iff" over all mismatches; a symmetry-only effect (families differ in parameter-to-function map and optimisation geometry)
Inference-time averaging helps the deployed predictor	E-S1: -0.0486, CI $[-0.0708, -0.0284]$, 7/12 eligible runs, graphs; E-S5b: same tool helps every arm (§7.2)	that the LLM found the group; coverage-conditional, not intention-to-treat
Verifier feedback improves search	**Not established**: primary difference undetected on both families, CIs wider than the prespecified equivalence margin	equivalence; "clean null" is not a finding we claim
Verification identifies the symmetry group	**Refuted as implemented**: 10/299 correct groups on graphs, 0/192 on ring; 289 wrong-group proposals pass label preservation	that no verifier could identify a group; our gate is a soundness filter
LLM search beats simple alternatives	**Not established**: constructive equivariant beats all arms on ring; edge-count threshold beats all arms on graphs	quantum advantage of any kind

contrast, coverage of the intervention, paired reliability endpoints, and verifier selectivity measured against the true group.

4. **Baselines that the LLM story has to beat** (§7.5): a constructive equivariant architecture at the same fit budget, a task-aware classical rule, the invariant oracle, and the same repair tool applied to non-LLM search.

2 Related work, re-verified

Every entry of §2 was checked against a primary record (arXiv API, Crossref, PMLR); all eight had at least one material error, and the corrected list is in the bibliography with the audit in `research/paper/REFERENCES_AUDIT.md`. One of my own round-1 audit conclusions was also wrong and is withdrawn in that file: I had recorded that [1] does not support invariant re-uploading, and re-reading the primary text shows it does. That source constructs "variational quantum data re-uploading models that make invariant predictions", defines an *equivariant embedding* ($U(V_s[x]) = U_s U(x) U_s^\dagger$), and gives the *twirling formula* $T_U[X] = |S|^{-1} \sum_s U_s X U_s^\dagger$ as a projector onto the commutant, applied to gates and whole ansätze. Our inference-time average is the output-side specialisation of that

projector, and our constructive equivariant baseline is an instance of their blueprint; neither is a contribution of ours.

The rest of the picture is equally established: equivariant unitary layers and channels with constructive prescriptions [2]; permutation-invariant QML for graph connectedness, directly occupied territory [3]; symmetry alone does not determine a useful architecture [4]; averaging-operator generalisation benefits, including projection identities of the kind we use [5]; enforcing, discovering and promoting symmetry by Lie-derivative methods [6]; finite-measurement task-level symmetry inference with projection bias, the closest conceptual neighbour [7]; and the LLM-evolutionary harness we build on is ShinkaEvolve [8]. The bandit ensemble that selects a proposer per generation is *not* from [8]: it is a modification made by our own prior study [9], and we attribute it there.

What is left, stated precisely. The averaging operator, the projection identity, the invariant-model blueprint and the label-preservation test are all prior art, and the limitation we document for the verifier (a validity test cannot distinguish a group from its valid subgroups) is a structural property of such tests, not a discovered fact about LLM reasoning. What this paper contributes is a *measurement* of the decision problem that sits between having a symmetry tool and using it: on identical fitted predictors, we quantify how much of the averaging benefit a label-preservation gate actually lets through (sensitivity 1/6 on the invariant target, 0/60 repairs fired on the ring, 0 false accepts), show that on one of the two task families the gate is *provably vacuous* before any model runs (the target's label-symmetry group is the whole hypothesis space), show that a verifier-free measured-defect gate captures the same benefit with no harm, and show that the negative control — averaging over a mismatched group — is where verification would earn its keep. That is a statement about when to apply a known repair, which the prior literature on how to build the repair does not settle.

3 Preliminaries: symmetry diagnostics and their assumptions

Let P_g be a specified unitary representation of a finite group G on $\mathbb{C}^d$ and U a circuit unitary at fixed numerical parameters. The *strict* residual

$$D_U(g) \;=\; \frac{\|U - P_g U P_g^\dagger\|_F^2}{2d} \in [0,2] \tag{1}$$

vanishes iff U commutes with P_g. For physical unitary channels we use the *channel* residual

$$D_{\text{chan}}(g) \;=\; 1 - \left|\mathrm{Tr}(U^\dagger P_g U P_g^\dagger)/d\right|^2 \in [0,1], \tag{2}$$

which vanishes exactly when the conjugation differs by a global phase (e.g. Z anticommutes with X: strict residual 2, channel residual 0). Adjacent transpositions generate S_n; zero residual on all generators certifies exact symmetry *at the sampled parameters*. A random-parameter zero is evidence about a family, not a proof over a family; where a proof exists we give it (Appendix A).

3.1 From block equivariance to predictor invariance

Block equivariance alone does not give an invariant predictor. Writing the expanded circuit as $V(x) = B_L\, U_{\text{enc}}(x) \cdots B_1\, U_{\text{enc}}(x)$ with readout M, $f(x) = \langle 0\cdots 0|V(x)^\dagger M V(x)|0\cdots 0\rangle$ is G-invariant when

1. **encoding equivariance:** $U_{\text{enc}}(gx) = P_g\, U_{\text{enc}}(x)\, P_g^\dagger$ (the ring encoding applies $\mathrm{RY}(x_i)$ on wire i, so a site permutation is implemented by P_g; the graph encoding applies $\mathrm{RZZ}_{\pi/2}$ on the edge indicated by bit k, and S_n acts on edges);
2. **invariant preparation:** $|0\cdots0\rangle$ is fixed by P_g;
3. **block equivariance:** each block commutes with P_g;
4. **invariant readout:** $[M, P_g] = 0$ for all g — we use the fixed readout $M = \frac{1}{n}\sum_i Z_i$ and do *not* train the readout, because a learned readout erases the bias.

Conditions (1), (2) and (4) are structural and checked once; condition (3) is what the residual measures. Because a residual on the block is not a certificate on the predictor, we additionally measure predictor invariance directly, on all group images of every evaluation row: with f fitted and $\mathcal{T}_G f$ its Reynolds average over the orbit, $\mathbb{E}\big[(f - \mathcal{T}_G f)^2\big]$ is reported per arm (§7.1, Table 5); it is 0 to roundoff for the repaired arm and $O(10^{-2})$ for the asymmetric arms.

3.2 The projection identity, and the floor, with their measures stated

The Reynolds operator $\mathcal{T}_G(U) = \frac{1}{|G|}\sum_g P_g U P_g^\dagger$ is an orthogonal projection for the Hilbert–Schmidt inner product, so the mean strict residual over a uniform group equals $\|U - \mathcal{T}_G(U)\|_F^2/d$. For predictions on inputs, let μ be the input distribution and let $\bar{\mu} = \frac{1}{|G|}\sum_g g_{\#}\mu$ be its orbit average. μ is *invariant* iff $\bar{\mu} = \mu$.

Proposition 1 (risk decomposition on an invariant input measure)**.** *Let μ be G-invariant, $y = h + \eta b$ with $\mathcal{T}_G h = h$ and $\mathcal{T}_G b = 0$, and $R(f) = \mathbb{E}_{x\sim\mu}(f(x) - y(x))^2$. Then for every f,*

$$R(f) \;=\; R(\mathcal{T}_G f) + \mathbb{E}\big[(f - \mathcal{T}_G f)^2\big] - 2\,\mathbb{E}\big[(f - \mathcal{T}_G f)(y - \mathcal{T}_G y)\big], \tag{3}$$

so the cross term vanishes when y is invariant ($\eta = 0$), giving $R(f) = R(\mathcal{T}_G f) + \mathbb{E}[(f - \mathcal{T}_G f)^2] \geq R(\mathcal{T}_G f)$, and for every invariant *f, $R(f) = \mathbb{E}[(f - h)^2] + \eta^2\mathbb{E}[b^2] \;\geq\; \eta^2\mathbb{E}_\mu[b^2]$.*

The proof is the orthogonality of the projection plus $\mathbb{E}[g\, b] = \mathbb{E}[g\, \mathcal{T}_G b] = 0$ for invariant g, which uses the invariance of μ essentially. The sign matters and an earlier draft of this paper had it wrong: with an invariant target the post-hoc projection can only *help* ($R(f) \geq R(\mathcal{T}_G f)$), but when the target breaks the symmetry the cross term dominates and $R(f) - R(\mathcal{T}_G f) < 0$ — measured, -0.013 to -0.018 at η=0.5 and -0.033 to -0.047 at η=1 for the asymmetric arms, i.e. projecting an asymmetric fit onto the invariant subspace *hurts* exactly where invariance is wrong. The full identity holds to 3.7×10^{-16} over all 2340 fits of WP-REPAIR-2 (5.5×10^{-18} at η=0). In our benchmark $\mu = \mathrm{Unif}[-1, 1]^6$ with the site-permutation action, a product of identical marginals, hence permutation-invariant, so the hypothesis holds. Three quantities must not be conflated, and the first version of this paper did conflate them:

- the **population floor** $\eta^2\mathbb{E}_\mu[b^2]$ (here η^2 exactly, because b is normalised on the training split);
- a **Monte-Carlo estimate** on sampled orbit representatives, which has sampling error (on our development split it is 0.866–1.059 across breaking directions at η=1, mean 0.948 — the first version of this paper quoted one direction's value, 0.866, as if it were the floor);
- the **exact finite-orbit floor** on the orbit-completed evaluation set (each representative replaced by all $|G|$ images, uniform weights), which is exact for the sampled orbits and is the Rao-Blackwellised version of the second.

A sample containing one representative per orbit admits invariant interpolants of its labels, so no non-vacuous empirical lower bound exists on the representative sample alone; that is why the orbit-completed set, with declared weights, is what we use for floor accounting.

4 Self-audit of the prior study

The audited study is *an earlier version of this project by the same author*: our prior study [9], archived in this repository as `README.md` at git commit `13ea13eb76af2f587c508aefde81b9dd026b3ce1` with run artefacts under `experiments/transfer-sn/` and `analysis/`. We state this explicitly because auditing one's own prior work changes what the audit can and cannot establish: we are competent to inspect the code, and we are also the party with an incentive to soften the findings. Four findings bound what the old numbers can mean; each is reproducible with `python3 research/audit.py`, which executes the historical evaluator's own abstract syntax tree rather than a paraphrase of it.

The prior study's separate single-run LLM reports (under `experiments/transfer-sn/`) are exploratory by their own account — one incomplete run with context-length failures, non-random missing generations, and a score that used train–test information — and are excluded from every claim here: no model ranking, effect size or generalisation statement in this paper depends on them, and they are never merged with the 120-run prospective study.

Test leakage in selection. The selection score used a train–*test* gap and the public feedback disclosed test accuracy and loss. The historical "test" set measured an adaptively reused development set and cannot establish held-out generalisation.

A refuted equivariance claim. The hand-designed baseline was claimed to be permutation-equivariant for all parameters. Its one-direction CRZ layer is not: at a documented parameter setting its mean adjacent-generator channel defect is 0.0597, and a bidirectional half-angle repair reduces it to 1.5×10^{-14}. A separate ordered-CRX counterexample shows that covering every unordered pair does not guarantee symmetry, because non-commuting controlled rotations are order dependent.

Non-independence. 563 selected proposals are not 563 independent experiments: they share ancestry, data and adaptive selection, yet significance was computed proposal-by-proposal.

A trivial shortcut. A single edge-count threshold fitted on training data alone attains 83.0% historical validation accuracy. In the fresh benchmark used here the same rule reaches *development* accuracy 91.2% and risk 0.351 (§7.5), which is better than every search arm at the same endpoint; density, not structure, carries much of the signal. Its risk on the same evaluation rows as the arms is reported in §7.8 so the comparison is matched.

5 WP-REPAIR: a causal repair experiment with explicit units

Targets. On the ring benchmark (inputs $x \in \mathbb{R}^6$, group D_6 of order 12 acting by site relabelling), h is the dataset's ground-truth *invariant* oracle $h(x) = \frac{1}{n}\sum_i x_i x_{i+1}$ and, for a frozen random direction v, $b(x) \propto \big(v\cdot x - \mathcal{T}_G(v\cdot x)\big)$ is the group-centred remainder of a linear functional, normalised so $\mathbb{E}_{\text{train}}[b^2] = 1$. Targets are $y_\eta = h + \eta b$, $\eta \in \{0, 0.5, 1\}$. h and b are functions of the *input*, never produced by a compared ansatz, and the Reynolds projection is computed by applying the explicit group closure to each input (the dataset's orbit labels are singletons and cannot be used).

Replication units (the fix). At $\eta = 0$ the target does not contain b, so the breaking direction is a nuisance index, not a replication: the unit is the *optimiser seed*. At $\eta > 0$ the unit is the (direction, optimiser seed) pair, with direction a blocking factor. Analysis code counts rows and units separately and refuses to resample rows: the first version's "24 paired draws" at $\eta = 0$ are 8 units written three times, and Table 2 reports what changes when they are collapsed. We use 8 directions and 24 optimiser seeds, so $\eta = 0$ has 24 genuine units. Three breaking directions cannot support a population claim about arbitrary target mismatches; eight support a within-benchmark statement, and the cluster bootstrap over directions is reported with that caveat.

Arms. All arms train three block parameters $\{a, b, c\}$ with the fixed symmetric readout of §3.1.

- `original`: one-direction CRZ, 15 gates, asymmetric;
- `repaired`: bidirectional shared-angle CRZ, 30 gates, exactly equivariant (analytically, Appendix A);
- `control_dup`: same-direction CRZ twice, 30 gates, asymmetric;
- `control_rand`: per-pair direction from a fixed seeded rule, 30 gates, asymmetric;
- `control_matched`: `control_dup` with the initial angle halved, so its initial *operator* is identical to `original`'s.

What equal gate counts do and do not buy. `control_dup` is the `original` interaction at angle $2b$ (verified to 10^{-14}, Appendix A), so it matches gate count and reachable function family but *not* optimisation geometry, and identical numerical initialisations do not match initial operators: the measured initial-operator distance is 1.37 for `control_dup` and 0 for `control_matched`. We therefore describe the contrast as "the effect of these particular circuit interventions under this training procedure" and never as a symmetry-only effect. Optimiser-budget sensitivity (200/600/1800 steps) is reported separately.

Function-space estimand. For each fitted f we also evaluate f and $\mathcal{T}_G f$ on the orbit-completed development distribution (uniform over 12 group images $\times$ representatives), reporting $R(f) - R(\mathcal{T}_G f)$, $\mathbb{E}[(f - \mathcal{T}_G f)^2]$ and the cross term of Proposition 1. This is a deterministic statement about one predictor; it is not a statement about retrained architectures, and we do not use the former as a theorem about the latter.

6 WP-SEARCH: estimands, endpoints and statistics

The frozen record is the 120-run confirmatory study: five arms (`random`, `mutation-ea`, `llm-fitness`, `llm-verify`, `llm-verify-repair`) at identical evaluated-candidate budget B=12, identical seed-derived initial populations, identical per-candidate training budget (Adam, 120 steps, 256 training rows), every invalid proposal retained in the denominator, one model treatment (*Qwen3.8-Flash-Next* at reasoning effort `none`; identity pinned in Appendix C), 12 paired seeds per family.

Endpoints, named. *Development risk* is risk on the validation split of the development artefact; that split is used by candidate selection, by the search feedback loop and by the verifier, so it is not held out in any confirmatory sense. *Sealed test risk* is read once, after the protocol and analysis

Table 2: The first version's repair record, recomputed with correct replication units. At η=0 the 24 rows are 8 units (the target is direction-free); the interval barely excludes 0 and neither p-value is small, so the claim "enforcing equivariance causally improves risk at η=0" is *not* supported by this record. At η>0 the units are (direction, seed) and the cost of enforced invariance is robust.

η	contrast	rows	units	mean diff	95% CI	p_{boot}	p_{flip}
0.0	repaired − control	24	**8**	−0.00093	[−0.00217, −0.00002]	0.083	0.125
0.0	repaired − original	24	**8**	−0.00201	[−0.00394, −0.00021]	0.034	0.125
0.0	control − original	24	**8**	−0.00107	[−0.00277, +0.00041]	0.191	0.320
0.5	repaired − control	24	24	+0.01817	[+0.01317, +0.02305]	$< 10^{-4}$	$< 10^{-4}$
0.5	control − original	24	24	−0.00127	[−0.00546, +0.00289]	0.549	0.564
1.0	repaired − control	24	24	**+0.04523**	[+0.03225, +0.05773]	$< 10^{-4}$	$< 10^{-4}$
1.0	control − original	24	24	−0.00204	[−0.01346, +0.00903]	0.721	0.729

code were committed. *Fresh holdout risk* (ring only) uses a holdout generated after the freeze with zero canonical-orbit overlap with any development or sealed-test row. The first version of this paper wrote "held-out risk" for development numbers; that wording is removed everywhere.

Estimands. E-S1: paired effect of inference-time Reynolds averaging on the *same* frozen dev-selected candidate, reported with coverage (eligible / attempted / fired) because it is coverage-conditional and not intention-to-treat. E-S2: the deployed pipeline (averaged predictor where the repair fired, candidate otherwise). E-S3 (primary): `llm-verify` minus `llm-fitness` development-selected risk. E-S4: paired reliability endpoints (invalid-proposal rate, hypothesis-present rate, verifier false-accept rate against the true group, nonconstant-output check). E-S5: development risk inside edge-count bands of `graphs_n6` — the intervention the "density shortcut" explanation requires. E-S5b: the same averaging tool applied to every arm's selected predictor using the task's *declared* group, i.e. a non-LLM pipeline given the same tool. E-S6: matched-budget baselines.

Statistics. Percentile bootstrap of the mean paired difference over *units* (20,000 resamples); p-values from the null-centred (shifted) bootstrap plus a sign-flip test, whose assumption — the null distribution of paired differences is symmetric about 0 — is stated wherever it is used; simulated p-values carry a floor and are never reported as 0; leave-one-unit-out ranges accompany every primary; Holm correction within declared families only (A: the two primaries; B/C: the repair contrasts at η=0/η=1; D: secondary search contrasts per family and endpoint; E: the breaking-target study). A confidence interval containing 0 is reported as "difference not detected", never as "no effect", "unchanged" or "clean null". Practical equivalence is claimed only when a whole interval lies inside the prespecified margin $\delta = 0.05\,\mathrm{Var}_{\text{dev}}(y)$ (0.03274 for graphs, 0.00097 for ring).

7 Results

7.1 Repair: corrected units change the headline

Table 2 recomputes the original repair record under the corrected replication units. The nothing in the η=0 family survives multiplicity (Holm p=0.166, 0.103, 0.191 for the three contrasts); the prose/table contradiction disappears (control minus original now includes 0, but that is "not detected",

Table 3: Floor accounting on the same record, with per-unit floors instead of one quoted number (the "floor" column is the mean over the 8/24 units; the repaired arm's per-unit excess ranges $[-0.0098, +0.0524]$ at η=1). The floor constrains *invariant* predictors only, so the asymmetric arms sit *below* it by fitting the breaking component; the repaired arm sits +0.023 above it (mean over units), which is why "pinned near the floor" is replaced by a number.

η	arm	mean dev risk	floor (mean over units)	excess over floor
0.0	original	0.01511	0.000	+0.01511
0.0	repaired	**0.01311**	0.000	+0.01311
0.0	control	0.01404	0.000	+0.01404
0.5	original	0.23842	0.23710	+0.00132
0.5	repaired	0.25532	0.23710	+0.01822
0.5	control	0.23715	0.23710	+0.00005
1.0	original	0.92796	0.94838	−0.02042
1.0	repaired	0.97114	0.94838	**+0.02276**
1.0	control	**0.92591**	0.94838	−0.02247

not equivalence); and the "iff" statement is withdrawn — what survives is that the mean differences have opposite signs at the tested mismatch levels, with the η=1 cost robust and the η=0 benefit not established.

What WP-REPAIR-2 decides. With the unit definition enforced and three asymmetric controls, the η=0 picture is: the repaired arm is better than the `original` (−0.00134, Holm p=0.043) and better than the random-direction control (−0.00171, Holm p=0.006), but the gate-count-matched duplicate control closes most of that gap (−0.00060, CI $[-0.00135, +0.00019]$), and the operator-matched control is indistinguishable from the original (+0.00003) — which is the validation that the matching works. The defensible statement is therefore: *under this training procedure, enforcing equivariance improves risk over the asymmetric architectures we tried by an amount whose symmetry-specific part is smaller than our resolution*, and the honest headline is the asymmetry: the cost at η>0 is 60× the size of the η=0 benefit and is detected at every budget. Table 5 shows why: the repaired arm's predictor is invariant to roundoff ($\mathbb{E}[(f - \mathcal{T}_G f)^2] \sim 10^{-33}$) while the asymmetric arms keep $O(10^{-3})$ of breaking component, which is worth 0.02 of risk at η=1.

Budget sensitivity. Repeating the principal contrast at 600 and 1800 Adam steps (fewer units: 6 at η=0, 12 at η>0) leaves the η=0 repaired-versus-duplicate-control difference undetected at every budget (−0.00135, CI $[-0.00316, +0.00006]$ at 600; −0.00111, CI $[-0.00288, +0.00014]$ at 1800), while the η=1 cost of enforced invariance *grows* with the budget (+0.036 at 200, +0.051 at 600, +0.057 at 1800 steps, all intervals excluding 0): more optimisation makes the asymmetric arms better at fitting the breaking component, which is the mechanism Proposition 1 predicts.

7.2 Search: the omitted endpoint is the one that works

Table 6 separates the two endpoints that the original study conflated. An earlier version reported `verify-repair` versus `verify` on the *unaveraged* candidate risk and concluded that an averaging benefit seen in a 3-seed pilot "did not survive" confirmation. Measured on the deployed predictor, the averaging effect on graphs is −0.0486 with an interval excluding 0 (sign-flip p=0.016), and the

Table 4: WP-REPAIR-2 with the unit definition enforced by design: η=0 is emitted once (the target is direction-free) with 24 optimiser seeds, η>0 has 8 directions × 24 seeds = 192 units. Holm is applied within the η=0 and η=1 families. Reading: the repaired arm does beat the `original` and the random-direction control at η=0, but the *gate-count-matched duplicate* control closes most of that gap, so the symmetry-specific part (−0.0006) is not established. At η>0 the cost of enforced invariance is robust and grows with the optimiser budget.

η	contrast	units	mean diff	95% CI	p_{boot}	p_{flip}	p_{Holm}
0.0	repaired − control_dup	24	−0.00060	[−0.00135, +0.00019]	0.123	0.150	0.246
0.0	repaired − control_rand	24	**−0.00171**	[−0.00276, −0.00080]	0.0012	0.0004	**0.006**
0.0	repaired − original	24	**−0.00134**	[−0.00237, −0.00031]	0.011	0.018	**0.043**
0.0	control_dup − original	24	−0.00074	[−0.00166, +0.00002]	0.082	0.133	0.246
0.0	control_matched − original	24	+0.00003	[−0.00033, +0.00053]	0.875	0.937	0.875
0.5	repaired − control_dup	192	+0.01142	[+0.00958, +0.01329]	$< 10^{-4}$	$< 10^{-5}$	$< 10^{-3}$
0.5	repaired − original	192	+0.00879	[+0.00719, +0.01043]	$< 10^{-4}$	$< 10^{-5}$	$< 10^{-3}$
0.5	control_dup − original	192	−0.00263	[−0.00457, −0.00069]	0.0083	0.0081	0.017
0.5	control_matched − original	192	−0.00043	[−0.00124, +0.00032]	0.276	0.295	0.276
1.0	repaired − control_dup	192	**+0.03605**	[+0.03134, +0.04087]	$< 10^{-4}$	$< 10^{-5}$	$< 10^{-3}$
1.0	repaired − control_rand	192	+0.02797	[+0.02296, +0.03316]	$< 10^{-4}$	$< 10^{-5}$	$< 10^{-3}$
1.0	repaired − original	192	+0.02526	[+0.02135, +0.02920]	$< 10^{-4}$	$< 10^{-5}$	$< 10^{-3}$
1.0	control_dup − original	192	−0.01080	[−0.01545, −0.00620]	$< 10^{-4}$	2×10^{-5}	$< 10^{-3}$
1.0	control_matched − original	192	−0.00234	[−0.00439, −0.00043]	0.021	0.021	0.021

deployed-pipeline contrast moves from +0.0009 (candidate endpoint) to −0.0274 (deployed endpoint). The primary hypothesis — that verifier *feedback* improves selected risk — remains undetected on both families, and neither interval is inside the prespecified equivalence margin, so we do not claim equivalence either. On the deployed endpoint the repair arm does beat the fitness arm (−0.04158, CI [−0.08572, −0.00566], shifted-bootstrap p=0.041), while on the candidate endpoint the same pair is undetected (−0.01322): the endpoint correction is what makes the arm comparison meaningful.

Coverage. On `graphs_n6`, 7 of 12 runs attempted a repair and all 7 fired; on `ring_n6`, 4 attempted and 0 fired (the label-preservation gate rejected them). E-S1 is therefore a per-protocol estimate; the arm-level intention-to-treat question is answered by E-S2, which is underpowered at 12 seeds.

The tool, not the LLM. Applying the same declared-group averaging to every arm's selected predictor (E-S5b) lowers development risk for *all* arms, including the non-LLM ones: graphs −0.0639 (`llm-fitness`) to −0.1114 (`random`); ring −0.0031 to −0.0039; all intervals exclude 0. Whatever the averaging buys, it does not require an LLM to buy it. **Approximation, stated plainly:** the declared group of `graphs_n6` is S_6 with 720 elements, so every *declared-group* twirl for graphs in this paper is a *Monte-Carlo* average over 24 sampled group elements, not the exact Reynolds projection — the MC share is 1.00 on graphs and 0.00 on the ring, where $|D_6| = 12$ is enumerated exactly. The deployed endpoint is the exception and must not be conflated with it: there the hypothesis groups are small subgroups (2–12 elements, all seven fired graph repairs included) enumerated *exactly*, so deployed graph risks are exact averages while the declared-group column is the 24-element approximation. Guarantees that hold for the exact projector (equivariance of the averaged predictor, the projection identity) do *not* transfer to a 24-element average; what we report for graphs is the measured risk change of that approximation, and its cost is 12.5 s per candidate against 0.18 s on

Table 5: WP-REPAIR-2 floor accounting on the orbit-completed distribution (exact finite-orbit floor, uniform weights over the 12 group images), with the measured predictor invariance $\mathbb{E}[(f - \mathcal{T}_G f)^2]$ and the mean channel residual. The repaired arm is exactly invariant (residual and asymmetry at roundoff, as proved in Appendix A); the asymmetric arms sit below the floor at η=1 by fitting the breaking component.

η	arm	dev risk	orbit risk	exact floor	excess	$\mathbb{E}[(f - \mathcal{T}_G f)^2]$
0.0	original	0.01471	0.01473	0.00000	+0.01473	1.4×10^{-4}
0.0	repaired	**0.01337**	0.01337	0.00000	+0.01337	9×10^{-34}
0.0	control_dup	0.01397	0.01399	0.00000	+0.01399	4×10^{-5}
0.0	control_matched	0.01475	0.01477	0.00000	+0.01477	1.8×10^{-4}
0.0	control_rand	0.01508	0.01509	0.00000	+0.01509	1.0×10^{-4}
0.5	original	0.24164	0.24612	0.24021	+0.00591	1.4×10^{-3}
0.5	repaired	0.25042	0.25587	0.24021	+0.01565	1.2×10^{-33}
0.5	control_dup	**0.23900**	0.24289	0.24021	+0.00268	2.0×10^{-3}
1.0	original	0.93566	0.95210	0.96085	−0.00875	1.8×10^{-3}
1.0	repaired	0.96091	0.98029	0.96085	**+0.01944**	1.6×10^{-33}
1.0	control_dup	**0.92486**	0.94043	0.96085	−0.02042	2.8×10^{-3}
1.0	control_matched	0.93332	0.95003	0.96085	−0.01082	2.0×10^{-3}
1.0	control_rand	0.93294	0.95224	0.96085	−0.00861	2.0×10^{-3}

Table 6: E-S3 (primary) and E-S1 (the endpoint the first analysis omitted). E-S1 is coverage-conditional: on `ring_n6` the repair never fired, so there is no E-S1 estimate there at all — an absence the first version did not report.

estimand	family	n	mean diff	95% CI	verdict
E-S3 verify − fitness	graphs	12	−0.01414	[−0.05855, +0.02837]	not detected
E-S3 verify − fitness	ring	12	+0.00058	[−0.00024, +0.00140]	not detected
E-S1 averaged − candidate	graphs	**7/12**	**−0.04862**	[−0.07083, −0.02839]	averaged better
E-S1 averaged − candidate	ring	**0/12**	no data (4 attempted, 0 fired)		
E-S2 deployed verify-repair − verify	graphs	12	−0.02744	[−0.05401, +0.00519]	not detected
E-S2 deployed verify-repair − fitness	graphs	12	−0.04158	[−0.08572, −0.00566]	repair arm better
E-S2 deployed verify-repair − verify	ring	12	−0.00046	[−0.00127, +0.00045]	not detected

the ring.

7.3 Verification does not identify the group

For every proposal that carried a hypothesis we recomputed, on the trusted side, its status against the declared group and against the measured label-symmetry group of the scored target (§7.10). On `graphs_n6`, of 299 proposals carrying a hypothesis, 10 are `exact_declared` (S_6) and 289 are `valid_proper_subgroup`: genuine label symmetries, weaker than declared, and **no** invalid hypothesis at all — which is not a surprise once $|L_t| = 720$ is measured, because on that target no permutation hypothesis can fail the test. On `ring_n6`, of 192 proposals, 0 are `exact_declared`, 120 are valid proper subgroups of D_6, and 72 are genuinely invalid for the scored target; the gate accepted 7 of them and rejected 95 that *were* symmetries of the scored target.

This is a structural property of validity tests, not an implementation bug and not a discovered fact

Table 7: Breaking-target selectivity test (development risk on each run's own validation split; refits reproduce logged risks to $\leq 7 \times 10^{-18}$). The gate's *accept* rate tracks label preservation (8 of 348 hypotheses pass here, against 299 of 299 on the invariant graphs target), so the filter works; identification still fails (0 of 348 carry D_6); repair never fires; and no arm ordering is detectable.

arm	dev risk	accuracy	hyp.	pass gate	= task group	repair fired
random	0.04122	0.9122	69	0	0	—
mutation-ea	0.04123	0.9153	79	0	0	—
llm-fitness	0.04049	0.9118	56	0	0	—
llm-verify	0.03699	0.9163	73	4	0	—
llm-verify-repair	0.04106	0.9170	71	4	0	0 of 6

paired contrasts (family E, Holm-adjusted)
verify − fitness -0.00350 $[-0.00963, +0.00198]$, p_{Holm}=0.74
verify-repair − verify $+0.00407$ $[-0.00069, +0.00984]$, p_{Holm}=0.67
verify-repair − fitness $+0.00058$ $[-0.00535, +0.00649]$, p_{Holm}=1.00
verify − random -0.00423 $[-0.01027, +0.00135]$, p_{Holm}=0.67
mutation-ea − random $+0.00001$ $[-0.00280, +0.00311]$, p_{Holm}=1.00

about LLM reasoning: testing that a proposed group preserves labels cannot distinguish a group from its valid subgroups, because every element of a subgroup of a label symmetry preserves labels by construction. A finite-sample label test is a *screen*, not a universal soundness certificate. Read correctly, the implemented gate is sound (measured false-accept rate 0/7 on the ring, 0/144 on graphs) and severely under-sensitive (it rejected 95 of 95 sound ring hypotheses, CI $[0.961, 1.000]$). Establishing *identification* of the maximal group would need a different instrument (counterexample search over the group, or a symmetry-breaking probe distribution), which we did not build; §7.10 measures what the existing gate does decide, and §7.11 measures what that costs.

7.4 Does the verifier's effect track the target's symmetry? (Addendum A1)

The selectivity test was declared in the protocol before it was run: all five arms on `ring_n6` with target `alpha_1` (the declared group is *not* a label symmetry of this target), 12 fresh paired seeds, its own Holm family, and the prediction stated in advance — if the verifier's benefit came from pushing the search toward the declared group, it must turn negative when that group is wrong.

Three things follow. The verifier's *filter* is selective: it accepts 100% of hypotheses when the target is invariant and 2.3% when it is not, which is the behaviour its definition promises. The verifier's *identification* is not improved: no proposal carries the task's group, and the repair step never fires. And the verifier's *effect on search* is undetectable in either direction (largest absolute mean difference 0.0042, all Holm $p \geq 0.67$), so the pre-declared prediction is not testable at this resolution: with the symmetry removed, verification neither measurably helped nor measurably hurt. That is consistent with the primary null — the feedback channel rarely changes the selected candidate — and inconsistent with any large "verification helps" story. For reference, moving from the invariant to the breaking target costs every arm ≈ 0.031 of risk (0.0065–0.0102 to 0.037–0.041), and no arm detects the difference.

Table 8: E-S7: sealed-test risk of the frozen 120-run *candidates*, refitted deterministically (reproducing logged development risk to 1.7×10^{-18}) and scored once. The sealed split reproduces the development pattern: on the ring the LLM arms beat the non-LLM arms and the verifier-versus-fitness difference is undetected; on graphs no arm difference is detected. Baselines are not compared across endpoints or splits here; the matched-row comparison is Table 10.

family	arm	dev risk	sealed-test risk	paired vs `random` on test
ring	random	0.01013	0.00995	—
ring	mutation-ea	0.01022	0.01025	+0.00030 [−0.00009, +0.00090]
ring	llm-fitness	0.00651	0.00639	**−0.00356** [−0.00469, −0.00239]
ring	llm-verify	0.00709	0.00711	**−0.00284** [−0.00442, −0.00117]
ring	llm-verify-repair	0.00664	0.00697	**−0.00298** [−0.00452, −0.00154]
graphs	random	0.53782	0.48704	—
graphs	mutation-ea	0.53930	0.49753	+0.01049 [−0.01082, +0.03816]
graphs	llm-fitness	0.56572	0.51806	+0.03102 [−0.04167, +0.09514]
graphs	llm-verify	0.55158	0.49272	+0.00568 [−0.04052, +0.05050]
graphs	llm-verify-repair	0.55250	0.50674	+0.01970 [−0.00448, +0.05037]

7.5 Baselines the LLM story has to beat

At the same fit budget (120 steps, 256 rows, 12 paired seeds), the hand-built equivariant architecture (bidirectional CRZ, measured channel residual 4×10^{-15}) reaches development risk 0.00055 on `ring_n6`; every search arm is worse, by +0.0060 (`llm-fitness`, CI [+0.0055, +0.0064]) to +0.0097 (`mutation-ea`), all $p < 10^{-4}$. On `graphs_n6` the same construction is *worse* than every arm (0.667 vs 0.538–0.566): at the search protocol's per-fit budget this exactly equivariant three-parameter block did not reach the arms' risk. We report that as an optimisation outcome at a fixed budget, *not* as an expressivity impossibility — no proof or exhaustive representational argument is offered — which is the two-sided lesson of [4]. The classical edge-count rule (threshold fitted on training rows only, k=6) reaches development risk 0.351 and accuracy 0.912 on graphs, beating every arm on that split; on matched held-out rows the advantage does not transfer (0.57812 vs the arms' 0.501–0.545, Table 10). The best constant predictor reaches 0.657; the invariant oracle on the ring is 0 by construction. Density stratification (E-S5) puts the arms' error where the shortcut lives: in the 0–5 edge band the five arms' mean risk is 1.439 against 0.167 in the 8–9 band, and the edge-count rule's band risks are 1.13 (0–5), 0.13 (6–7), 0.00 (8–9) and 0.05 (10–11) — the same profile, which is why we describe the graphs task as shortcut-dominated rather than claiming the shortcut *caused* the null without an intervention that changes arm ordering.

7.6 One-shot sealed evaluation of the search arms (E-S7, candidate endpoint)

Every development-selected candidate was refitted deterministically (reproducing its logged development risk to 1.7×10^{-18}) and scored once on the sealed split, with the file digests recorded before and after (`research/HOLDOUT_ACCESS.json`). This is the *candidate* endpoint: it scores the circuit the search selected, before the deployment averaging step, which is measured separately in §7.8.

The two prespecified contrasts of interest behave as on development: `llm-verify` − `llm-fitness` on the ring is +0.00072 (CI [−0.00039, +0.00186], not detected) and on graphs −0.02534 (CI [−0.11142, +0.05299], not detected), while `llm-fitness` − `mutation-ea` on the ring is −0.00386 (CI [−0.00491, −0.00277]). Test risks are uniformly a little below development risks on graphs (the

Table 9: E-R4' on untouched inputs (4 breaking directions × 8 optimiser seeds: 8 units at η=0, 32 at η>0), next to the same unit subset on development rows. The untouched data reproduce the WP-REPAIR-2 reading: the repaired arm beats the original architecture, the gate-count-matched duplicate control is *not* beaten at η=0, and enforcing invariance costs 0.046 of risk at η=1. Magnitudes differ slightly from Table 4 because this is a subset of directions and seeds.

	contrast	development (same units)		untouched inputs	
η		mean diff	95% CI	mean diff	95% CI
0.0	repaired − control_dup	−0.00083	[−0.00225, +0.00024]	−0.00057	[−0.00176, +0.00028]
0.0	repaired − original	**−0.00233**	[−0.00417, −0.00057]	**−0.00174**	[−0.00309, −0.00051]
0.0	control_dup − original	−0.00150	[−0.00351, +0.00024]	−0.00117	[−0.00279, +0.00015]
0.5	repaired − control_dup	+0.01757	[+0.01326, +0.02180]	+0.01551	[+0.01106, +0.02011]
0.5	repaired − original	+0.01265	[+0.00875, +0.01658]	+0.01153	[+0.00804, +0.01501]
0.5	control_dup − original	−0.00492	[−0.00955, −0.00033]	−0.00399	[−0.00853, +0.00043]
1.0	repaired − control_dup	+0.05138	[+0.04083, +0.06123]	**+0.04629**	[+0.03613, +0.05670]
1.0	repaired − original	+0.03440	[+0.02521, +0.04384]	+0.03067	[+0.02227, +0.03906]
1.0	control_dup − original	−0.01697	[−0.02978, −0.00450]	−0.01561	[−0.02814, −0.00358]

sealed split is easier), so the level of the development numbers is not an artifact of the split — but no arm difference on graphs survives the move to sealed data, which is the third time this section has reported a development effect that does not materialise.

7.7 Untouched-input evaluation of the repair arms (E-R4')

The first one-shot attempt at this evaluation was *invalid* and is reported as such: the evaluator scored every arm against a single invariant label array per split, so at η>0 it answered a different question (the tell was a risk of 0.019 at η=1 where the invariant floor is ≈ 1). Protocol Addendum A4 records the defect, keeps the two ring splits it had already read consumed, and replaces them with an **inputs-only** holdout (seed 20261204, 4000 rows, zero dihedral-orbit overlap with development or any earlier test split, sha256 recorded, read once), evaluated only after a gate that refits every WP-REPAIR-2 unit and must reproduce the experiment's own development risk exactly (`216/216` compared, max difference 0.0).

The untouched-input result is the cleanest statement this paper can make about repair: on data no part of the pipeline touched, the difference between enforced equivariance and a gate-matched asymmetric control is *not detected* when the target is symmetric (−0.00057, CI [−0.00176, +0.00028]; we do not claim the two are equal), and enforcing it costs 4.6% of the target variance when the target is not. The file digests before and after the read are identical (`research/HOLDOUT_ACCESS_v4.json`), and the evaluator now refuses to open the two splits consumed by the defective run.

7.8 The deployed endpoint on held-out inputs (A5.5)

Held-out search numbers for the selected *candidate* circuit are in §7.6; they do not evaluate the pipeline this paper emphasises — averaging at inference time — so this subsection measures that pipeline on held-out rows, with the development self-check running *before* any held-out array is opened (120/120 refits reproduce their logged development risk, max 1.73×10^{-18}). Three endpoints are scored on one shared 1024-row subsample per family (seed 5150), so every comparison below is

Table 10: Deployed-endpoint evaluation on held-out rows (means over the 12 frozen runs per arm, identical rows within a family). "Repair fired" counts the development-frozen deployed decision. Deployed hypothesis groups (2–12 elements) are enumerated exactly; the declared-group column for graphs is a 24-element Monte-Carlo average of a 720-element group, while the ring declared group (12) is exact.

family	arm	candidate	deployed	declared-group twirl	fired
graphs	random	0.50548	0.50548	**0.40915**	0/12
graphs	mutation-ea	0.50895	0.50895	0.42886	0/12
graphs	llm-fitness	0.54472	0.54472	0.47449	0/12
graphs	llm-verify	0.50941	0.50941	0.43511	0/12
graphs	llm-verify-repair	0.52913	**0.50067**	0.44353	7/12
ring	random	0.01036	0.01036	**0.00578**	0/12
ring	mutation-ea	0.01052	0.01052	0.00592	0/12
ring	llm-fitness	0.00648	0.00648	**0.00311**	0/12
ring	llm-verify	0.00712	0.00712	0.00348	0/12
ring	llm-verify-repair	0.00676	0.00676	0.00319	0/12
graphs	edge-count rule (k=6, train-fitted)	—	0.57812	—	—
graphs	constructive equivariant	—	0.59936	—	—
graphs	best constant	—	0.58968	—	—
ring	constructive equivariant	—	**0.00026**	—	—
ring	best constant	—	0.01946	—	—

on identical rows: the *candidate* (as before), the *deployed* predictor (candidate averaged over the group its development-frozen repair decision selected), and the *declared-group twirl* applied to every arm including `random`.

Two data facts are part of the result. `ring_n6` uses genuinely fresh inputs (v5, seed 20261205, dihedral-orbit-disjoint from development and from splits v1–v4, first read, digests logged). For `graphs_n6` a first-and-only read is *impossible*: exhaustive enumeration of all 2^{15} labeled six-vertex graphs leaves 0 isomorphism classes unused by development and splits v1–v4, so the graphs numbers below are a *second* read of the v1 split, labelled post hoc, authorised explicitly in the run command and recorded in the consumption ledger.

Repair on the endpoint that matters. Deployed averaging changes the risk only where the frozen decision fires: `llm-verify-repair` on graphs, 0.52913 → 0.50067 (−0.02846, CI [−0.04704, −0.01190], 7/12 runs). Everywhere else deployed equals candidate exactly. On the ring that equality is itself a result: the deployed gate fires on 0/12 runs on *fresh* inputs. Because the evaluator uses the development-*frozen* decision, that zero is guaranteed by the evaluation design: fresh inputs validate the *risks* of the frozen policies and cannot test the firing mechanism, which requires newly fitted candidates (§7.9 refits candidates under corrected gates).

The tool still dominates the arm. Applying the declared group to every arm beats every arm's own selection: `random` with the twirl reaches 0.40915 on graphs, below all five deployed arm risks (0.50067–0.54472), and 0.00578 on the ring, below the best arm's candidate (0.00648). The constructive equivariant block reaches 0.00026 on the *fresh* ring inputs — 25× below the best arm's candidate and 12× below the best arm's own twirl. Held-out data does not rescue the search from either comparison.

Table 11: Corrected vs defective gate on identical frozen ring candidates (60). Δ is risk after − risk before (negative helps); "realised" counts a non-firing candidate as 0; costs are measured seconds per candidate. On `graphs_n6` the mismatched and approximate regimes do not exist (no mismatched group exists), and in the valid regimes every gate fires on every candidate, so no rule differs there.

regime	rule	fired	realised Δ	always	harm rejected	benefit missed	gate s
valid (D_6)	R0 always	60/60	−0.00345	−0.00345	0/0	0/60	0.0000
	R1 defective	0/60	0.00000	−0.00345	0/0	60/60	0.0015
	R1 corrected	**60/60**	**−0.00345**	−0.00345	0/0	0/60	**0.0017**
	R3 defect (q_1)	45/60	−0.00275	−0.00345	0/0	15/60	0.0755
	R_rand75	45/60	−0.00254	−0.00345	0/0	15/60	0.0000
mismatched	R0 always	60/60	+0.00266	+0.00266	0/58	0/2	0.0000
	R1 defective	0/60	0.00000	+0.00266	58/58	2/2	0.0015
	R1 corrected	0/60	**0.00000**	+0.00266	**58/58**	2/2	**0.0005**
	R3 defect (q_1)	45/60	+0.00209	+0.00266	15/58	0/2	0.0189
	R_rand75	45/60	+0.00208	+0.00266	14/58	1/2	0.0000
ϵ=10^{-12}	R1 corrected	60/60	−0.00345	−0.00345	0/0	0/60	0.0017
ϵ=10^{-9}	R1 corrected	0/60	0.00000	−0.00345	0/0	60/60	0.0017
ϵ=10^{-6}	R1 corrected	0/60	0.00000	−0.00345	0/0	60/60	0.0017
ϵ=10^{-3}	R1 corrected	0/60	0.00000	−0.00344	0/0	60/60	0.0017

A development-only baseline advantage does not transfer. The edge-count rule reaches development risk 0.351 against the arms' 0.538–0.566 (§7.5); that comparison alone invites the claim that it "beats every arm". Scored on the *same* held-out rows as the arms it does not: 0.57812 against 0.50067–0.54472. Its development advantage does not transfer. The splits also differ in label composition — the best-constant reference moves from 0.65667 on development validation to 0.58968 here — so risk *levels* are not comparable across splits, which is why every table in this paper scores arms and baselines on one shared row set.

7.9 The corrected gate is the comparator that matters (A6)

The two defects above mean the preceding section describes *our implementation*, not verification in general. So we built the corrected screen exactly as declared in Addendum A6.1 (scored target only; a violation means $|y_t(gx) - y_t(x)| > 10^{-12} + 10^{-9}|y_t(x)|$, the probe's own tolerance, so the gate cannot be stricter than the instrument that defines soundness), re-ran the whole decision comparison on the *same* 120 frozen predictors, and added the regimes that make the decision non-trivial: the supplied group is the declared group (*valid*), a valid proper subgroup, a *mismatched* group that is not a label symmetry, and the declared group under label perturbations $\epsilon \in \{10^{-12}, 10^{-9}, 10^{-6}, 10^{-3}\}$. Every rule is scored under *its own* firing mask in every regime; costs are measured, including the cost of the defect statistic, which needs the averaged predictions on its own sample.

Four results, none of which was available before the corrected comparator existed.

(1) The recall bottleneck on the valid group was the bug, not the gate. The corrected screen accepts all 60 ring candidates and captures the entire benefit (−0.00345, identical to unconditional averaging) for 0.0017 s of gate time. The implemented screen accepts none. Any statement of the form "verification is a recall bottleneck" is therefore withdrawn: it described a screen that tested the wrong labels at zero tolerance.

(2) The gate does have a safety role, and now we have measured it. Under a mismatched group, averaging harms 58 of 60 candidates (mean +0.00266). Both screens reject all 58; the defect statistic rejects 15 and a size-matched random rule 14, so both let roughly three quarters of the harm through. Under the cost objective of A6.3 (J = risk $+w\cdot$(gate + one-off averaging + H per-prediction overhead)), the corrected label gate dominates unconditional averaging, the defect rule and the random rule at every (H, w) considered in this regime, while on the valid group it is itself dominated by averaging — which is the honest answer to "is a verifier worth it": *only* for the risk that the supplied group is wrong, and its value depends on how often that risk is real.

(3) A strict label gate forfeits benefit under approximate symmetry, and the tolerance *is* the policy. With the frozen tolerance, the corrected gate accepts at $\epsilon = 10^{-12}$ and rejects at every larger perturbation, while averaging still helps by -0.00345 at all of them. The crossover sits exactly at the tolerance, so the choice of tolerance is not an implementation detail but the decision boundary; a sweep over $\{0, 10^{-15}, 10^{-12}, 10^{-9}, 10^{-6}, 10^{-3}\}$ is reported so no single frozen value is load-bearing. This is a limitation of *any* exact label test, not of our code.

(4) The defect rule is neither cheap nor much better than random. Its statistic costs 0.0755 s on the ring against 0.0017 s for the corrected label test (44×) and 3.66 s against 1.39 s on graphs, because it needs averaged predictions on its own sample — an accounting this module makes explicit because the statistic is not free. Retaining 75% of candidates by this statistic yields -0.00275 against -0.00254 for a size-matched random selection: a difference of 0.00021, which we do not present as a method. It is also not verifier-free in the earlier sense: it measures predictive symmetry, it simply omits the label test.

A structural point about gate metrics. In every regime the label test's decision is *constant across candidates* (it tests the supplied group against the target on gate rows; nothing about the candidate enters). Precision and recall of a pure label gate over candidates are therefore degenerate by construction — 45/45 precision for a rule that never looks at the candidate is not evidence about the rule. Only the channel and defect tests vary per candidate. We report per-arm and seed-vector results in the generated record and reserve interval language for quantities with a defensible sampling unit.

7.10 What a label-preservation verifier can decide (A5.2, A5.3)

A hypothesis closure H is classified against the declared group G and the *measured* label-symmetry group L_t of the scored target (every one of the $n!$ site permutations tested on an independent probe panel of 512 development rows that the gate never reads). Counting $H \neq G$ as a wrong group was wrong: $H < G$ is a genuine symmetry that is merely weaker than declared.

Two consequences. First, on `graphs_n6` the label gate is *vacuous by construction*: connectedness is invariant under every relabelling of the vertices, so no permutation hypothesis can violate it, and the observed 144/144 acceptances say nothing about the quality of the proposals. The 289 hypotheses classified as `valid_proper_subgroup` are sound, incomplete, and **zero** false accepts. Second, on the ring the scored target α_0 admits 12 of 720 permutations, but the implemented gate checks *every* target column in the dataset, whose conjunction admits only 2 of 720 — six times stricter than the scored target requires.

Table 12: Measured label-symmetry groups. $|L_t|/|S_n|$ is the fraction of *individual uniformly drawn permutations* that the target's labels preserve, under an exact membership test and a known true L_t. It is *not* the acceptance probability of a search proposal, which is a generator list whose closure is sampled non-uniformly; that quantity is $\Pr_{H\sim Q}[H \leq L_t]$ and is estimated separately (Table 13). $|L_t|/|S_n| = 1$ still entails vacuity for *any* proposal distribution supported on permutations of this domain, which is the only claim this table carries.

family	target	$\lvert L_t\rvert$	$\lvert L_t\rvert/\lvert S_n\rvert$	contains G	$= S_n$
graphs	connected	720	1.0000	yes	**yes**
graphs	triangle	720	1.0000	yes	**yes**
graphs	ALL	720	1.0000	yes	yes
ring	alpha_0	12	0.0167	yes	no
ring	alpha_0.1 / 0.25 / 0.5 / 1	2	0.0028	no	no
ring	ALL	2	0.0028	no	no

Table 13: Acceptance probability of the proposal distribution that matters, $\Pr_{H\sim Q}[H \leq L_t]$, by 400 independent Monte-Carlo draws per cell (Wilson intervals are defensible here because the draws are independent by construction, which the fixed panel was not). The "uniform model" column is Pr under the idealised Q the text describes: $|L_t|/|S_n|$ for one uniform permutation and $(|L_t|/|S_n|)^k$ for k independent ones. The empirical row draws the generator lists actually proposed by the model in the frozen search runs (819 hypotheses). At $n = 9$ the 362880 permutations are never enumerated: the estimate is probe-based, and the algebraic anchor (the eight square symmetries, all sound on the probe) gives $|L|/|S_9| = 2.2 \times 10^{-5}$.

Q	k	ring α_0 accept	95% CI	uniform model (idealised)	graphs accept	tictactoe accept
uniform single	1	0.0225	[0.0119, 0.0422]	0.01667	1.0000	0.0000
uniform k	2	0.0025	[0.0004, 0.0140]	0.00028	1.0000	0.0000
uniform k	3	0.0000	[0.0000, 0.0095]	0.0000046	1.0000	0.0000
empirical LLM	1	**0.4700**	[0.4216, 0.5190]	—	1.0000	—

The uniform models are right and irrelevant. An independent 4000-draw check gives single-generator soundness 0.0179 and two-generator soundness 0.00025, matching $|L_t|/|S_n|$ and $(|L_t|/|S_n|)^k$. But the proposals the model actually made accept at 0.4700: 21× the uniform single-permutation rate and roughly 1700× the two-generator model. A diagnostic that ignores Q is off by three orders of magnitude on the only Q this paper studies, which is why we now report the distribution-aware quantity and treat the ratio as a special case of it. The two degenerate ends are measured rather than asserted: graphs accepts under every Q (no predicate on this domain can discriminate), and the $n = 9$ board task rejects under every Q (a gate that rejects everything discriminates nothing either). A useful gate needs Q to put mass on both sides of L_t.

The sensitivity loss has two separable causes, each measured: the all-columns conjunction, and a zero tolerance applied to a continuous target. The declared rotation is a genuine symmetry of α_0 — its worst violated fraction on the probe is 0.0 at the probe's 10^{-9} tolerance — yet at a tolerance of exactly 0 the gate records a violated fraction of 0.53125 across the rotation's closure (34 of its 64 gate rows) and rejects it. With both causes corrected, panel sensitivity on the invariant target goes from 1/6 to 6/6 with no false accepts. On the search-generated hypotheses the recorded decisions give 0 false accepts out of 7 accepted on the ring (CI $[0.000, 0.354]$) and 95 of 95 rejected

Table 14: Fixed-panel selectivity (A5.3, units corrected by A6.6): identical ring inputs, only the target varies. The panel is *nine generator sets*, fixed in advance (identity, rotation, reflection, the declared dihedral group, its C_3 and C_2 subgroups, the full symmetric group, 20 random permutations as one joint generator list, and 15 single transpositions as one joint generator list). These are *not* independent draws from a population of transformations, so no interval is attached: counts are descriptive coverage, and the distinct-subgroup column is given next to the entry count because the last three entries all generate S_6 (seven distinct subgroups among nine entries; the full listing is in `research/search/corrected_gate.json`).

target	gate variant	sound accepted (entries)	of distinct sound subgroups	false accepts	$\lvert L_t \rvert$
α_0	implemented (all columns, tol. 0)	1/6	1/6	0/3	12
α_0	scored target, tol. 0	1/6	1/6	0/3	12
α_0	scored target, tol. 10^{-9}	**6/6**	**6/6**	0/3	12
α_1	implemented	1/2	1/2	0/7	2
connected	implemented	9/9	7/7	0/0 (no violator exists)	720

hypotheses *sound* for the scored target (CI $[0.961, 1.000]$): the implemented gate's failure mode is missed sound hypotheses, not accepted unsound ones. Recovering D_6 on a target that is deliberately not D_6-invariant would be the wrong answer, and we no longer score it as a failure.

7.11 Verification as a decision rule (A5.4)

That declared-group averaging helps non-LLM pipelines does not show a verifier is necessary, so we stopped asserting it and measured the decision instead. For each of the 120 frozen candidates we applied the same averaging and scored the same rows under five rules. The outcome is paired per candidate (Δ = risk after − risk before; negative helps), and a rule's *realised* change counts a non-firing candidate as 0.

Four things follow. (i) Averaging over the supplied group never hurt a single one of the 120 candidates, so on these targets unconditional averaging is the better policy and the verifier is *not* a precondition for the benefit. (ii) The implemented gate forfeits 100% of the available benefit on the ring (realised 0.00000 against −0.00345), which is the operational cost of the sensitivity loss measured above. (iii) A verifier-free gate on the predictor's own measured non-invariance captures 79% of it (−0.00274) at precision 45/45 with no candidate harmed. (iv) The negative control is where verification would earn its keep: averaging over a mismatched group costs +0.00399, so a rule that fires blindly is unsafe exactly on targets whose symmetry differs from the supplied group. Two further mechanical facts matter for design: the channel condition in the deployed gate is satisfied by 60/60 candidates in both families, so it contributes no discrimination, and on graphs the label test costs 0.45 s while the averaging it gates costs 12.5 s — the filter is 3.6% of the work it is supposed to save, and it filters nothing.

8 Discussion

What the evidence supports. (a) Enforcing equivariance has a large, robust *cost* when the target breaks the symmetry: +0.03605, CI $[+0.02998, +0.04210]$ under the primary estimand (fixed direction set, resampled optimiser seeds), rising to +0.057 at 1800 optimiser steps; the direction-

Table 15: Decision-rule comparison on the 120 frozen candidates (60 per family), for the gate *as implemented* — the corrected gate is compared separately in §7.9 and changes the reading. Precision is the share of fired repairs that helped; recall the share of beneficial repairs that fired. The final row is *not* a rule: it averages the mismatched-group risk change over every candidate with no firing mask, so it is an unconditional stress test and not any gate's realised risk. It is undefined on graphs because no mismatched group exists there.

family	rule	fired	realised Δ	if always applied	precision	missed benef
graphs	R0 supplied group (always)	60/60	−0.08705	−0.08705	60/60	
graphs	R1 label gate	60/60	−0.08705	−0.08705	60/60	
graphs	R2 deployed gate	60/60	−0.08705	−0.08705	60/60	
graphs	R3 defect gate (q_1)	45/60	−0.07544	−0.08705	45/45	
graphs	unconditional mismatch stress test	—	undefined: L_t is the whole space			
ring	R0 supplied group (always)	60/60	−0.00345	−0.00345	60/60	
ring	R1 label gate	**0/60**	**0.00000**	−0.00345	0/0	
ring	R2 deployed gate	**0/60**	**0.00000**	−0.00345	0/0	
ring	R3 defect gate (q_1)	45/60	−0.00274	−0.00345	45/45	
ring	unconditional mismatch stress test (no mask)	n/a	+0.00399	—	—	

cluster, crossed and legacy cell-level resamplings give $[+0.02610, +0.04597]$, $[+0.02366, +0.04828]$ and $[+0.03134, +0.04087]$, and all four exclude 0. (b) At η=0 a benefit that the gate-count-matched control removes: the repaired arm beats the original architecture but not the duplicate-gate control, so no symmetry-specific benefit is established. (c) The analytic repair — inference-time averaging over a supplied group — is the only intervention in this study that improves the deployed predictor consistently (-0.04862 on development rows, -0.02846 on held-out rows), and it does so for non-LLM pipelines too. (d) Verification is worth studying as a *decision*: as implemented, the label screen contained two specification defects, and once they are fixed the screen captures the full benefit on a valid group (-0.00345 at 0.0017 s of gate time), rejects 58 of 58 harmful repairs under a mismatched group, and forfeits the benefit entirely once the target is only approximately symmetric — with the crossover sitting exactly at the numerical tolerance. (e) At matched budgets a hand-built equivariant block and a one-threshold classical rule beat LLM search on development splits, and the equivariant block still dominates on fresh held-out ring inputs (0.00026 against the best arm's 0.00648); the classical rule's development advantage does not survive matched held-out rows.

What that implies for the "verification helps automated science" thesis. Verification helped in exactly one place, and it is a *safety* place: it is what stops a wrong supplied group from being averaged over. It did not help the model find the group, the feedback channel we built left development-selected risk with a difference we cannot detect, and it is not a precondition for the repair benefit — averaging over a group we supply helps whether or not any gate checks it. That reframes the design question. A label-preservation screen should be judged the way one judges a fuse: by the harm it prevents, at the frequency the hazard occurs, relative to its cost — not by whether it improves average performance on a benchmark where the supplied group happens to be right. On these tasks that frequency is 0 on `graphs_n6` (no mismatched group exists) and 1 of 1 on the ring's mismatched regime, and the same screen that is free insurance in the second case is pure overhead in the first.

What would change these conclusions. Two things, both expensive and neither performed here. First, matched search runs with the corrected feedback: post-processing frozen candidates cannot recover the trajectories that corrected feedback would have produced, so no behavioural claim about LLM feedback in this paper is stronger than "difference not detected". Second, an independent task family whose target is approximately rather than exactly symmetric, which is the regime where the choice of tolerance, rather than the presence of a verifier, decides the outcome.

Relation to the pilot. A 48-run pilot (3 seeds, 3 treatments) preceded the 120-run study and is retained as exploratory. Its variance check set the confirmatory seed count. Dated git evidence for the pilot's protocol is not available (Appendix C), so we describe the search study as protocol-consistent rather than verifiably pre-registered; the repair-2, dev-analysis and one-shot holdout chain is verifiably pre-registered (protocol commit `ecd1be72`, 16:20:02Z, precedes job submissions at 17:10Z and 17:21Z).

9 Limitations

Small dense statevector simulation; two six-qubit development families plus one sealed ring holdout, with `tictactoe_n9` left sealed and unclaimed because dense n=9 simulation exceeds this revision's budget; a single model (*Qwen3.8-Flash-Next*), so no model comparison is claimed; budget B=12; three (now eight) breaking directions, which supports within-benchmark statements only; equal gate counts do not make two circuit families equivalent in expressivity or optimisation geometry, so no symmetry-only attribution is offered; the verifier's label gate has no power against subgroups of the true label symmetry, and on `graphs_n6` no power against anything, since its target's label-symmetry group is the whole hypothesis space; the *graphs* benchmark cannot supply a fresh test distribution at all — all 2^{15} labeled six-vertex graphs, hence all isomorphism classes, are already consumed by development and the earlier holdouts, so the graphs deployed-endpoint result is a labelled second read rather than untouched data; every *declared-group* `graphs_n6` average in this paper is a 24-element Monte-Carlo twirl of a 720-element group (deployed hypothesis subgroups of order 2–12 are exact), so exact-projector guarantees do not transfer to it; the defect gate in the decision-rule ablation is one statistic on one dataset per family, so it is a measured comparison on this benchmark and not a validated method; all repair inference conditions on a single dataset and task family, so no interval here covers newly sampled datasets or families; the model serving alias exposes no immutable checkpoint revision, so reproducibility of the LLM component is temporal rather than bit-level; and no quantum-advantage claim is made or implied.

10 Reproducibility and integrity

All splits are orbit-disjoint; the sealed splits were opened once for outcomes after the protocol freeze, with a metadata-only access incident recorded in `research/PROVENANCE.md` and remedied by a fresh holdout generated after the freeze; every result JSON records seeds, module versions, git commit and host (with two recorded defects in the older records); analysis code, run records, dataset manifests, the frozen protocol, the bibliography audit and the per-table commands are released with the paper. Negative and null results are reported as "not detected" with their intervals, and equivalence is claimed only against a prespecified margin.

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A CRZ algebra behind the arms

With the simulator's convention (q_0 most significant, $\mathrm{CRZ}_{ij}(b) = |0\rangle\langle 0|_i \otimes I_j + |1\rangle\langle 1|_i \otimes \mathrm{RZ}(b)_j$),

$$\mathrm{CRZ}_{ij}(b)\,\mathrm{CRZ}_{ji}(b) = \exp\!\big(-\tfrac{ib}{4}(Z_i + Z_j)\big)\,\exp\!\big(+\tfrac{ib}{2} Z_i Z_j\big), \tag{4}$$

$$\mathrm{CRZ}_{ij}(b)^2 = \mathrm{CRZ}_{ij}(2b) = \exp\!\big(-\tfrac{ib}{2} Z_j\big)\,\exp\!\big(+\tfrac{ib}{2} Z_i Z_j\big). \tag{5}$$

Both are verified numerically to 10^{-13} in `research/experiments/repair/test_repair2.py`. Equation (4) is the analytic reason the repaired arm is equivariant: the ZZ term is manifestly symmetric and the field term is a permutation-symmetric product of identical single-qubit rotations, so the block commutes with *every* permutation, not only D_6; the measured residual (4×10^{-16}) is a check, not the proof. Equation (5) is the analytic reason the gate-count-matched control is not: its field term hits only the target wire, so local fields are non-uniform. It also shows the control is the `original` interaction at angle $2b$, which is why an operator-matched control (`control_matched`, initial b halved) is needed and why the two controls are reported separately.

B Reproduction

Tests (all pass):

```
python3 -m unittest discover -s research/core      -p 'test_*.py'
python3 -m unittest discover -s research/experiments/repair -p 'test_*.py'
python3 -m unittest discover -s research/search    -p 'test_*.py'
```

Experiments and analyses (one command per table):

```
python3 -m research.experiments.repair.repair_experiment2 --workers 48 \
  --out research/experiments/repair/repair2_results.json
python3 -m research.experiments.repair.analyze_repair \
  --in  research/experiments/repair/repair_results.json \
  --out research/experiments/repair/repair_v1_reanalysis.json
python3 -m research.experiments.repair.analyze_repair \
  --in  research/experiments/repair/repair2_results.json \
  --out research/experiments/repair/repair2_analysis.json
python3 -m research.search.analyze2 --dir logs/runs_confirm \
  --out research/search/confirm_analysis2.json --label confirmatory
python3 -m research.search.dev_analysis --dir logs/runs_confirm \
  --out research/search/dev_analysis.json --workers 24
python3 -m research.search.analyze_breaking --dir logs/runs_breaking \
  --out research/search/breaking_target.json
python3 -m research.search.verify_semantics --runs logs/runs_confirm \
  --out research/search/verify_semantics.json
sbatch research/search/job_decision.sh          # A5.4 decision rules
python3 -m research.holdout_oneshot --splits dev --skip-search \
  --selfcheck research/experiments/repair/repair2_results.json   # gate, no sealed read
python3 -m research.holdout_oneshot --splits fresh_v4 --skip-search \
  --selfcheck research/experiments/repair/repair2_results.json \
  --out research/HOLDOUT_RESULTS_v4.json \
  --access-log research/HOLDOUT_ACCESS_v4.json   # ONE shot on v4 only
python3 -m research.holdout_search_contrasts     # E-S7 vs-random contrasts
DRY_RUN=1 sbatch research/job_deployed.sh        # A5.5 gate first, opens nothing
sbatch research/job_deployed.sh                  # A5.5 deployed endpoint, ONE read
python3 -m research.compile_results2 --out research/results_summary2.md
python3 -m research.paper.check_numbers          # manuscript numbers must match records
python3 -m research.build_manifest --out research/MANIFEST.md
```

Each one-shot command refuses a split recorded as consumed, so the commands above are safe to copy: re-running them cannot turn a consumed split into a second look. The mapping from every table in this paper to its code revision, result file, digest, endpoint, split, seed and status is in `research/MANIFEST.md`, which is itself generated.

What the digests prove, and what they do not. Recording a file's sha256 before and after a read shows the bytes were not modified while we read them. It does *not* show that the file had never been opened before, and neither `0444` permissions nor the evaluator's path check are access isolation: they are procedural guards against our own drift, and the consumed-split ledger in `research/MANIFEST.md` is the record that matters. Likewise, git gives a documented chronology — the commit that last touched an analysis file, and the commit that added the protocol — but it is not an independently certified preregistration service, and we do not describe it as

one. Stack: `Python/3.12.3`, `SciPy-bundle/2024.05` (NumPy 1.26.4, SciPy 1.13.1); LLM arms additionally `module load ycrc-llm/1.0`. CPU batch jobs run on the cluster's `day` partition with `OPENBLAS_NUM_THREADS=1`; the work is CPU-only, so no GPU partition is used (holding a GPU idle to satisfy old protocol wording would violate cluster utilization policy).

C Provenance

Model identity: the alias `ycrc-local` resolves to *Qwen3.8-Flash-Next*, root `/apps/software/openwebui_models/Qwen3.8-Flash-Next-FP8`, served by vLLM with `max_model_len` 262144; requests send `model` and `reasoning_effort="none"` only (no temperature, top_p or max_tokens), with a 180 s timeout and up to 8 retries (none needed: 584/584 calls succeeded on the first attempt, mean latency 2.75 s). Budgets for the 120-run study: 584 calls, 1,294,840 tokens (829,452 prompt / 465,388 completion), 1448 proposals of which 8 were invalid and retained, 18.1 core-hours. The gateway publishes no immutable checkpoint revision, so the pin is alias + root path + module version + date.

Two provenance defects in our own earlier records, recorded rather than repaired silently: (1) the run JSONs report `git_commit=a37fc2d6`, which does not contain the code that produced them (it was uncommitted at run time); (2) the plan document that claims to have been frozen before any outcome was seen first appears in git *after* the pilot and confirmatory runs, so the search study is described as protocol-consistent, not verifiably pre-registered. Full dated chain, sealed-file digests and the access incident: `research/PROVENANCE.md`.

D Change history and audit trail

The scientific narrative above is written once, in the present tense. Because the manuscript went through three review rounds, the record of what changed and why lives here instead, so a reader can identify the current estimand and claim without reconstructing the review history. Each row names the artifact that carries the evidence; none of it is required to follow the argument.

Two invariants held throughout: no held-out split was read twice without an explicit, logged authorisation (one such authorisation exists, for `graphs_n6`, because no fresh class-disjoint graphs inputs exist), and every number in the manuscript is regenerated from result JSONs by `research/compile_results2.py`. The lint (`research/paper/check_numbers.py`) confirms that each manuscript token appears in the generated record; the claim binder (`research/paper/claim_bind.py`) additionally binds headline values to result keys with their arm, endpoint and split in the surrounding context, and reports its own mutation-detection rates. Neither is a proof of correctness, and both are described by what they can and cannot detect.

Table 16: What changed under review, with the artifact that documents it. Withdrawn claims are listed because a reader who saw an earlier version needs to know which sentences no longer hold.

round	change
1	Replication units corrected (24 cells were 8 units); orbit-completed floors; fresh-input repair evaluation; one-shot holdout policy; projection-identity sign error found and corrected (verified to 3.7×10^{-16} over 2340 fits); references re-checked against primary records. `REVISED_PROTOCOL.md` `ecd1be72`, A3, A4.
1 → withdrawn	"Test-split" numbers from the original evaluator (shown to be an adaptively reused development set); the "pinned near the invariant floor" phrasing; the 0.866 pooled floor.
2	Deployed endpoint added on held-out rows; hypothesis taxonomy replaced an equality test that counted sound subgroups as false accepts; fixed-panel selectivity; decision-rule ablation; four estimands for the crossed design; numeric-token lint added; five reporting bugs fixed (reversed verdict wording, CI-only verdicts, 0.000 p-floors, mislabelled identity columns, equality phrasing). A5.
2 → withdrawn	"Verification is a precondition for repair"; "289 wrong-group proposals passed the gate"; "a classical rule beats every arm on graphs" (true on development, not on matched held-out rows); the claim that our citation audit was itself reliable; "independent replication" of a frozen decision.
3	Corrected gate as first-class comparator; per-rule mismatch masks (the previous mismatched-group row applied no mask); measured costs including the defect statistic's own averaging; approximate-symmetry regimes; panel counts reported as descriptive coverage over 9 generator sets (7 distinct subgroups on the n=6 families; 6 identities established at n=9, where three entries exceed the enumeration cap) instead of binomial trials; the group-size ratio replaced by $\Pr_{H\sim Q}[H \leq L_t]$; "wider by construction" removed; claim binding with mutation testing. A6.
3 → withdrawn	"The gate is a recall bottleneck" (it was the implementation); the group-size ratio as a predictor of gate behaviour; "verifier-free" for the defect rule; the numeric checker described as validation rather than lint.